

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2019
URP 2251
(Landscape Planning and Design)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is Landscape? Distinguish between “Landscape Feature” and “Landscape Element.” (10)
(b) Describe with example the importance of studying Landscape Planning and Design for the students of URP. (15)
(c) Draw the use of different types of line in a landscape plan. (10)
2. (a) Discuss the color schemes used in Landscape Planning and Design. (10)
(b) Discuss different types of texture with proper drawings. (10)
(c) What is Ecology? Describe the significance of ecology on human being. (15)
3. (a) “Order is an important landscape principle for achieving a sound landscape” - Explain the statement. (10)
(b) Explain the design scenario of front entry and utility of a residential landscaping with necessary sketch. (15)
(c) Discuss Nitrogen cycle with necessary diagram. (10)
4. Write short notes of the following (any 5): (35)
(a) Visual Weight
(b) Landscape with shrub and ground cover
(c) Repetition of Landscaping
(d) The Biosphere
(e) Asymmetrical Balance
(f) Landscape with native flowers and grasses
(g) Proportion in landscape principle

Section – B

5. (a) What is landscape site? (05)
(b) Explain the site inventory and analysis process. (15)
(c) What is climate and context? What are the basic elements of context? (10)
(d) What are the characteristics of Millennial Landscape Planning and Design movement? (05)
6. (a) Mention eight functions of trees and plants in eight sentences for Landscape Planning and Design. (10)
(b) What are the principles of Waterwise Landscaping? (10)
(c) Explain with sketch: (15)
(i) Pool (ii) Cascade (iii) Dock
7. (a) What is micro-climate? (05)
(b) How waterbody could be a micro-climatic solution for a residential plot? Explain. (10)
(c) Discuss the Landscape Planning and Design process of a plot for residential use. (20)
8. (a) Explain the city vibration pattern in terms of zoning with necessary sketches. (15)
(b) What is buffering in Landscape Planning and Design for urban areas? Explain with examples and sketches. (08)
(c) Explain with sketch: (12)
(i) T-square pattern (ii) Ribbon Pattern (iii) Axis (iv) Vista

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 2nd Year 2nd Term Regular Examination, 2019

Hum 2217

(Public and Local Government Finance)

Full Marks: 210

Time: 3 hrs.

- N.B.** i) Answer any THREE questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) Discuss the importance of studying public finance in context of Urban and Regional Planning. (15)
(b) Define public goods and common resources and give an example of each (20)
why do government try to limit the use of common resources?
2. (a) What is Negative Externality? Give an example of Negative Externality and a Positive Externality. Explain why market outcomes are inefficient in the presence of these externalities. (15)
(b) Greater consumptions of alcohol leads to more motor vehicle accidents and, (20)
thus, imposes cost on people who do not drink and drive. Illustrate the market for alcohol, labeling the demand curve, the social value curve, the supply curve, the social cost curve, the market equilibrium, and efficient level of output.
3. (a) What is meant by the efficiency of a Tax System? What can make a tax system inefficient? (15)
(b) Define Price Ceiling and Price Floor and give an example of each. Which leads to a shortage? Which leads to a surplus? Why? (20)
4. (a) What are the factors on which incidence of tax depends? Trace the incidence of estimate duty and inheritance tax. (15)
(b) Draw the supply and demand curves for cookies. If the government imposes a tax on cookies, show what happens to the price paid by buyers, the price received by sellers, and the quantity sold. In your diagram, show the deadweight loss from tax. Explain the meaning of the deadweight loss. (20)

Section – B

5. (a) Distinguish between the burden of Internal and External Public Debt. Can any generation shift the burden of public debt to future generation? (20)
(b) What are the objectives of Public Expenditure? How far does the public expenditure in Bangladesh fulfill those objectives? (15)
6. (a) Explain the role of deficit financing in lower – middle class income country (LMIC) like Bangladesh. (15)
(b) What is Decentralization? Explain the importance of decentralization for a country. (20)
7. (a) Distinguish between Balanced Budget and Deficit Budget. (10)
(b) What are the main characteristics of an Excellent Budget? (10)
(c) Discuss “Budget Cycle” to prepare a budget. (15)
8. (a) Discuss local government structure in Bangladesh. (15)
(b) Discuss role of City Corporation for Urbanization. (10)
(c) What are the sources of revenue of Union Parishad? Explain. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 2nd Year 2nd Term Regular Examination, 2019

CE 2241

(Principles of Environmental Engineering)

Full Marks: 210

Time: 3 hrs.

- N.B.** i) Answer any THREE questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) What are the objectives of Water Supply? Discuss briefly the elements of Water Supply. (13)
(b) Describe the Hydrological Cycle with neat sketches. (12)
(c) A 150 mm diameter tube well produces 100 l/s with a drawdown of 3 m and circle of influence of 120 m in diameter. The static depth of water in the well is 40 m. Calculate the coefficient of permeability of the aquifer in which the tube well is sunk. (10)
2. (a) State the method of Population Estimations. Which method do you consider best for Population Estimation in Bangladesh and why? (08)
(b) Write down the sources, standard values and significance of the following water quality parameters: (12)
(i) Chloride (ii) Hardness (iii) Iron (iv) Arsenic
(c) What are the main purposes of transmission and distribution systems? (05)
(d) How can water demand of a community be estimated? Mention the water consumption data and the corresponding recommended peak factor for the design of water supply systems in Bangladesh. (10)
3. (a) What are the objectives of Water Purification? What are the factors influencing settling velocity of particles in water? (06)
(b) Explain the process of Coagulation – Flocculation. How does it differ from plain sedimentation? (10)
(c) Explain the process of disinfection by Chlorination. (09)
(d) Calculate the dimension of rectangular settling tank to treat 100 m³ of water per hour, where the overflow rate is 0.75 m/hr and the detention time is 2 hours. (10)
4. (a) What is filtration of water? Describe briefly the theories of filtration. (12)
(b) Explain the operating difficulties of Rapid Sand Filtration. How can these difficulties be removed? (08)
(c) Explain in brief the processes of Arsenic and Hardness removal from Groundwater. (15)

Section – B

5. (a) What is Sanitation System? Describe the interrelationships between Water, Sanitation and Health Education. (10)
- (b) Distinguish between On - Site and Off - Site Sanitation System. Describe the suitability of Sanitation System based on Water Consumption and Population Density. (10)
- (c) Design a leach pit for both single and alternating twin off- set pit pour flush latrines serving a family of 20 members living in a rural area. Wastewater flow is 40 lpcd and soil is clay type. (15)
6. (a) What are the advantages and disadvantages of VIP Latrines? Distinguish between Compost and ROEC. (10)
- (b) What is Septic Tank System? Describe the processes in Septic Tank System. (10)
- (c) A Septic Tank is to be designed to serve a residential building having 24 residences. Wastewater generation rate is 120 lpcd. The tank is to be desludged every four years and minimum mean hydraulic detention time for sedimentation is 2 days. (15)
7. (a) Distinguish between Separate and Combined Sewerage System. What are Self- Cleansing and Non- Scouring Velocity? (10)
- (b) Describe the technical advantage of SBS system over the Conventional Sewerage. (10)
- (c) Describe bacterial growth with neat sketch in waste water. Briefly describe the symbiosis between Algae and Bacteria. What are the advantages and disadvantages of Waste Stabilization Ponds? (15)
8. (a) Describe interrelationship between functional elements of Solid Waste Management. (10)
- (b) What is Composting? Show the basic composting process with a flow diagram. (10)
- (c) Compare the merits and demerits of hauled container and stationary container system of Solid Waste Collection. Which method do you think would be suitable for Khulna City? (10)
- (d) Describe the techniques to reduce Air Pollution. (05)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 2nd Term Regular Examination, 2019.
URP 2291
(Statistics for Planners – II)

Full Marks: 210

Time: 3.0hrs

- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.
iii) t-table, z-table, χ^2 , F-table.
iv) Assume data/values wherever necessary.

Section – A

1. (a) What are the key characteristics to identify a normal distribution? Explain in short. (08)
- (b) "The first central moment is always 0 and the second central moment equals to the variance"- explain. (06)
- (c) What is the difference between Raw Moment and Central Moment? Calculate the four Central Moments for the following data: 6,9,8,10,12,15. (13)
- (d) Graphically explain the concepts of Skewness and Kurtosis. (08)
2. (a) Explain the 68-95-99.7 rule in short. (07)
- (b) A scientific study compares the income (normal) distribution of farmers in Khulna and Sylhet district. It finds the following statistics: (23)

District	Khulna \$ US	Sylhet \$ US
Data		
Mean ($\bar{x}$)	369.84	301.63
Standard Deviation (σ)	382.81	301.02

Source: Alamgir et al. (2018)

Find the following:

- What percent of Khulna farmers belong to 752.65-1135.45 \$US income range?
 - What is the two income range of Sylhet for 2σ ?
 - What is your inference from the income distributions of the two districts?
- (c) In a normal distribution, at what standard deviation will you be able to cover 100% of a given data set? – explain. (05)
 3. (a) How do you decide whether to conduct parametric and non-parametric tests based on variables? Make a list. Mention the advantages and disadvantages of non-parametric tests. (10)
 - (b) "Working hypothesis is the basis of all scientific study" – explain. (05)
 - (c) A department head of KUET claims that his/her department's students have on an average higher CGPA than others. A random sample of 35 students of his/her department shows an average CGPA of 3.33. If the average CGPA of 850 KUET is 3.25 with a standard deviation of 0.10, is there a justification to his/her claim? (08)
 - (d) A statistics teacher of URP Batch 2k17 wants to compare the final examination mark for course URP-2191 (Statistics for Planner-I) and URP-2291 (Statistics for Planners-II) for a class of 60. Course URP-2191 has an average score of 70 and standard deviation of 17. Course URP-2291 has an average score of 65 and standard deviation of 12. Using $\alpha=0.05$, have the students performed differently between the two courses? Perform appropriate test following standard statistical procedure. (12)

4. (a) What is degree of freedom? (03) (03)
 (b) A medicine company is trying to test the effect of a drug prescribed for lowering Systolic Blood Pressure. The company tests blood pressure twice: before taking the medicine and one week after taking the medicine and records the following data. (12)

Cases	Before	After
1	135	127
2	142	145
3	137	131
4	122	125
5	147	132
6	151	147
7	131	119
8	117	125
9	154	132
10	143	139
11	133	122

Is the drug effective lowering systolic blood pressure? Explain with appropriate test.

- (c) What is the basic difference between Paired t-test and Anova test? Explain the concepts of 'between difference' and 'within difference' with examples. (10)
 (d) A Union Parishad Chairman is supposed to distribute humanitarian aid among four flood affected villages equally (based on population). The post aid distribution accounted for the following data: (10)

Village	A	B	C	D
Population	1200	1500	1450	980
% of aid receiver	24	22	25	29

Can you say that the aid distribution by the Chairman is unbiased?

Section – B

5. (a) "Correlation does not necessarily impose causation" – Justify the statement with appropriate examples. (10)
 (b) Make a relative comparison among the methods used to study correlation. (12)
 (c) A company gives on-the-job training to its salesmen which is followed by a test. It is considering whether it should terminate the services of any salesman who does not do well in the test. The following data give the test scores and sales made by 9 (nine) salesmen during the last one year. (13)

Test scores	14	19	24	21	26	22	15	20	19
Sales (BDT'000)	31	36	48	37	50	45	33	41	39

Compute the coefficient of correlation between test scores and sales. Does it indicate that termination of the services of salesman with low test score is justified?

6. (a) Illustrate the distinction between Correlation and Regression in brief. (12)
 (b) A panel of men and a panel of women were asked by a consumer testing organisation to rank 8 brands of tea according to taste. A rank of 1 was given to the best tasting tea and a rank of 8 to the worst. (12)

Brand	A	B	C	D	E	F	G	H
Panel of Women	5	4	3	6	7	8	1	2
Panel of Men	4	5	6	3	8	7	2	1

Determine how closely men's and women's tastes in tea are related.

- (c) The coefficient of rank correlation between debenture prices and share prices is found to be 0.143. If the sums of squares of the differences in rank is given to be 48, find the value of N. (11)

7. (a) Mention the properties of Regression Coefficients. Also, prove that Regression Co-efficients are independent of change of origin but not scale. (10)
- (b) Illustrate coefficient of Determination (R^2) with examples. Under what conditions can there be one regression line? Explain. (10)
- (c) You are given the following data about the sales and advertisement expenditure of a firm: (15)

	Sales (BDT. Crores)	Advertisement expenditure (BDT. Crores)
Arithmetic Mean	50	10
Standard deviation	10	2

Coefficient of correlation = + 0.9

- (i) Calculate the two regression equations
- (ii) Estimate the likely sales when advertisement expenditure is 13.3 crores (BDT.)
- (iii) What should be the advertisement budget if the company wants to achieve a sales target of BDT 70 crores?
8. (a) What is time series analysis? Illustrate the role of time series analysis in the decision making process of a planning project. (10)
- (b) Narrate the components of time series with necessary diagrams. (10)
- (c) The sales of a company in Lakhs of BDT for the year 1999 to 2005 are given below: (15)

Years	1999	2000	2001	2002	2003	2004	2005
Sales (in tonnes)	32	47	65	92	132	190	275

Estimate the sales figure for the year 2010 using an equation of the form $Y = ab^x$.